TOPIC 13

The P-value

The smallest significance level at which the null hypothesis can still be rejected, and how to use it for a decision

WHAT YOU WILL LEARN

- Define the p-value and explain why it was introduced
- Find a p-value from a z-score, one-sided and two-sided
- Decide by comparing the p-value with α at 10%, 5% and 1%
- Use the z-table when the exact value is not listed
- Know that p-values work for every test and distribution
- Report p-values correctly and avoid the usual mistakes

NOTE Where this connects

Topic 10 introduced hypothesis testing and Topic 12 the two errors a test can make. In both we fixed a significance level α in advance and asked a yes/no question. We tested at several levels (say 10%, 5%, 1%) but never found *where* the decision flips. The p-value answers exactly that.

1. What we know, and what we do not

We already know how to…	We do not yet know…
Test a hypothesis, and reject H_0 at a chosen level such as 5% or 1%	The level of significance **beyond which we can no longer reject** H_0

Trying levels one by one is clumsy. It is better to let the data tell us the exact boundary. That boundary is the **p-value**.

DEFINITION p-value

- **Class definition:** the smallest level of significance at which we can still reject the null hypothesis, given the observed sample statistic.
- **Equivalent standard definition:** the probability, **if H_0 is true**, of getting a result at least as extreme as the one observed.
- A small p-value means the data would be surprising if H_0 were true.

The two definitions agree. If the p-value is 0.034, then at $\alpha = 0.05$ we reject ($0.034 < 0.05$) and at $\alpha = 0.03$ we do not ($0.034 > 0.03$). 0.034 is the point where the decision changes.

2. Worked example 1: the data scientist salary

This is the example used in the class and in the confidence-interval topic. Salaries have a known population standard deviation $\sigma = 15{,}000$. A sample of $n = 30$ has mean 100,200. We test the published claim H_0: $\mu = 113{,}000$ against H_1: $\mu \neq 113{,}000$.

1. Standard error: $SE = 15{,}000/\sqrt{30} =$ **2,739** (the slide shows 2,739).
2. Test statistic: $z = (100{,}200 - 113{,}000)/2{,}738.6 =$ **−4.67**.

3. Look up the tail area in the z-table. Most printed tables stop at about z = 3.5 to 4, and the normal curve is symmetric, so −4.67 gives the same tail as +4.67. The nearest entry is about 0.0001 (the slide's 'p-value = 0.0001'). Read it as 'p is less than 0.0001'.
4. Exact value by computer: one tail = 1.48e-06; two-sided p = 2 × tail = **3.0e-06**.

Level α	Compare	Decision
10%	p ≈ 0.0001 < 0.1	Reject H_0
5%	p ≈ 0.0001 < 0.05	Reject H_0
1%	p ≈ 0.0001 < 0.01	Reject H_0

We reject at 10%, 5% and 1%, and in fact at every level people ever use. That is the 'extremely significant' case: the sample mean is almost five standard errors from the claimed value.

> **COMMON MISTAKE** **Why not just write 0.000?**
> The true p-value is tiny but never exactly zero. Software prints 0.000 when the value rounds to zero in three decimals. Write **p < 0.001** (here p ≈ 3e-06), not 'p = 0'.

3. Worked example 2: a result that splits the levels

Suppose another test gives z = 2.12. Now the decision depends on the level.

1. In the z-table, row 2.1 and column 0.02 give **0.9830** (about 0.983).
2. **One-sided** p-value = 1 − 0.983 = **0.017**.
3. **Two-sided** p-value = 2 × (1 − 0.983) = **0.034**.

Level α	Two-sided p = 0.034	Decision
10%	0.034 < 0.10	Reject H_0
5%	0.034 < 0.05	Reject H_0
1%	0.034 > 0.01	**Cannot** reject H_0

So z = 2.12 rejects at 5% but not at 1%, as the lecture says. The p-value, 0.034, shows where the switch happens: any α above 0.034 rejects, any α below it does not.



Figure 1. The p-value is the shaded tail area beyond the observed z. One-sided 0.0170, two-sided 0.0340.

Exact values by computer: one-sided 0.0170, two-sided 0.0340, the same as the table method.

4. The decision rule

KEY IDEA **Reject H_0 if p-value < α**

- **p < α:** reject H_0. The result is statistically significant at level α.
- **p ≥ α:** do not reject H_0 (the evidence is not strong enough). This is *not* proof that H_0 is true.
- The class summary ('p > 0.01 means accept H_0') is shorthand; the careful wording is *fail to reject*.

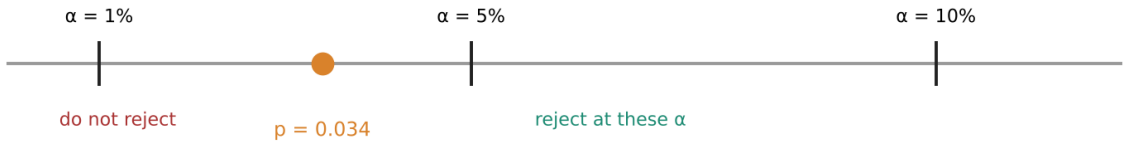


Figure 2. The p-value on the α scale. Every α to the right of p = 0.034 leads to rejection; every α to the left does not.

One-sided or two-sided?

A **two-sided** test counts results that are extreme in **either** direction, so the tail area is taken on both sides and, for a symmetric curve, doubled. Use it when a difference in either direction would matter (H_1: $\mu \neq \mu_0$), which is the safe default. Use a **one-sided** test only if, before seeing the data, only one direction matters (H_1: $\mu > \mu_0$ or $\mu < \mu_0$). The same z gives a one-sided p exactly half of the two-sided p.

Negative z: the normal curve is symmetric, so $P(Z \leq -z) = P(Z \geq z) = 1 - P(Z \leq z)$. That is why $z = -4.67$ and $z = +4.67$ give the same tail, and why you can look up the positive value in the table.

Who chooses α, and when?

The class notes say the researcher can decide the level 'post-factum', and that is one real advantage of p-values: the reader can compare your p with **their own** α. But you must not look at p first and then pick the α that gives the answer you want. State α **before** the analysis (Topic 10), then report the actual p-value alongside it.

Loose guide for strength of evidence

p-value	Informal reading
$p \geq 0.10$	Little or no evidence against H_0
$0.05 \leq p < 0.10$	Weak or borderline evidence
$0.01 \leq p < 0.05$	Moderate evidence
$0.001 \leq p < 0.01$	Strong evidence
$p < 0.001$	Very strong evidence

These bands are conventions, not rules of nature. Different fields use different thresholds.

5. The p-value is universal

The lecture stresses that the p-value is 'a universal concept': it works with **every test and every distribution**. Only the test statistic and its reference distribution change (t, chi-square and F are introduced in Topics 03, 09 and 11). The decision rule never does.

Test	Statistic	Reference distribution	scipy tail function
z-test	z	Standard normal	stats.norm.sf
t-test	t	t with df	stats.t.sf
Chi-square test	χ^2	Chi-square with df	stats.chi2.sf
ANOVA, regression F	F	F with (df_1, df_2)	stats.f.sf

In each case the p-value is the area in the tail(s) beyond the observed statistic. Most software computes it for you. You will meet it again in the regression table, where each coefficient has its own p-value (see the Regression Table topic of the regression course).

Observed result	p-value (computed)
t = 2.5 with 19 df, two-sided	0.0217
χ^2 = 7.8 with 3 df	0.0503
F = 4.2 with (2, 27) df	0.0258

6. What the p-value is not

Common mistake	Correct reading
'p = 0.034 means H_0 has a 3.4% chance of being true.'	The p-value assumes H_0 is true; it does not give the probability of H_0.
'A smaller p means a bigger effect.'	p depends on effect **and** sample size. Report an effect size or interval too.
'p > 0.05 shows there is no effect.'	It shows the data are not strong enough evidence. Power may be low (Topic 12).
'p = 0.000.'	Write p < 0.001.

Topic 10 covers these in more detail, including multiple testing.

7. In Python

```
import numpy as np
from scipy import stats

# example 1: salary
se = 15000 / np.sqrt(30)
z = (100200 - 113000) / se
print(round(z, 2), 2 * stats.norm.sf(abs(z)))        # -4.67  ~3e-06

# example 2: z = 2.12
print(stats.norm.cdf(2.12))                          # 0.983
print(stats.norm.sf(2.12), 2 * stats.norm.sf(2.12))  # one- and two-sided

# any test: p-value from its statistic
print(2 * stats.t.sf(2.5, df=19))                    # t-test, two-sided
print(stats.chi2.sf(7.8, df=3))                      # chi-square
```

SUMMARY

- The p-value is the smallest α at which H_0 can still be rejected; equivalently, P(result at least this extreme | H_0 true).
- Compute the test statistic, then take the tail area: one tail for one-sided, doubled for two-sided.
- Reject H_0 when $p < \alpha$; otherwise fail to reject.
- $z = -4.67$ gives $p \approx 3 \times 10^{-6}$ (table: 0.0001), rejecting at 10%, 5% and 1%.
- $z = 2.12$ gives a two-sided $p = 0.034$: reject at 10% and 5% but not at 1%.
- The p-value works for any test. Report it as $p < 0.001$ rather than 0.000, and never read it as the probability that H_0 is true.

Practice questions

Try each question before reading the answer underneath it.

Q1. A z-test gives $z = 1.64$ (one-sided, right tail). Find p and decide at $\alpha = 0.05$.

Answer. $p = P(Z \geq 1.64) = 0.0505 \approx 0.050$, just above 0.05 (borderline). At exactly 1.64 you do not reject at 5%; $z = 1.645$ is the cut-off.

Q2. For $z = -1.50$ in a two-sided test, find the p-value and decide at 5% and 10%.

Answer. $p = 2 \times P(Z \leq -1.50) = 0.1336$. $0.134 > 0.10$ and > 0.05, so fail to reject at both levels.

Q3. A software output gives $p = 0.000$. How should it be reported?

Answer. As $p < 0.001$. The value is tiny but not exactly zero.

Q4. Two researchers study the same data. One uses $\alpha = 0.05$ and rejects; the other uses $\alpha = 0.01$ and does not. Is that possible?

Answer. Yes. If the p-value lies between 0.01 and 0.05 (for example 0.034), it rejects at 5% but not at 1%. That is why p-values are reported.

Q5. Is 'p = 0.03 means there is a 97% chance the alternative is true' correct?

Answer. No. The p-value is calculated assuming H_0 is true; it is not the probability that either hypothesis is true.

Q6. Which is larger for the same data: the two-sided or the one-sided p-value (when the result is in the predicted direction)?

Answer. The two-sided p-value, exactly double for a symmetric distribution.